MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER, TYPES 6283 AND 8500

This amendment forms a part of MIL-E-1/1314B, dated 1 February 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Qualification inspection and quality conformance inspection, parts 1, 2, and 3: For tube type 6283 (only), the inspection criteria specified in this amendment supersedes and replaces those requirements specified in revision B of this document.

Method	Test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
---	Qualification							
	Life test	F = 1050 ±50 MHz; Ef = Note 2; Eb = 5000 V dc; Ec2 = 1000 V dc; Ec1 = -200 V dc; pd = 1.5 kW; tp = 5-15 µs; Du = 0.01 (see note 4)	---	---	t	500	---	hours
---	Life test end points	Power output F = 405-450 MHz; Ef = 6.3 V; Eb = 5000 V dc; Ec2 = 1000 v; Ec1 = -50 V dc; pd = 2.2 kw; tp = 15-17 µs; Du = .005 (see note 4)	---	---	Po	12.0	---	kw
1231		Pulsing emission sinusoid (see note 3)	---	---	is	40	---	a
	Quality conformance inspection, part 1							
1331	Heater current		0.65	II	If	3.5	4.0	A
1266	Total grid current	Ec1,1b = 300 mA dc; Eb = 2000 V dc; Ec2 = 300 V dc; (see note 7)	0.65	II	Ic1	---	-10.0	µA dc

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Method	Test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
	Quality conformance inspection, part 1 - continued							
1261	Electrode voltage (grid) (cutoff)	Ec2 = 300 V dc; Ec1/Ib = 10 mA dc	0.65	II	Ec1	-23.0	-34.0	V dc
1231	Pulsing emission sinusoid	ec1 = ec2 = eb = 750 v; tk = 90 (see note 3)	0.65	II	is	60.0	---	a
2214	Power output	F = 405-450 Mhz; Eb = 5000 V dc; ec2 = 1000 v; Ec1 = -50 V dc; pd = 2.2 kw; tp = 15-17 µs; Du = .005 (see note 4)	0.65	II	Po	12.0	---	kw
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	No voltage	---	---	---	---	---	---
1261	Electrode voltage (1) (grid)	Ec2 = 300 V dc; Ib = 200 mA	---	---	Ec1	-8	-15	V dc
1261	Electrode voltage (2) (grid)	Ec2 = 250 V dc; Ib = 200 mA	---	---	Ec1	-5	-11	V dc
1261	Electrode voltage (3) (grid)	Ec2 = 300 V dc; Ib = 250 mA	---	---	Ec1	-5	-12	V dc
1316	Amplification factor	(See note 5)	---	---	Mu	11	17	
1306	Transconductance	(See note 6)	---	---	Sm	20,000	---	µmhos
1331	Direct-interelectrode capacitance	g1 to g2 (anode grounded)	---	---	Cin	17	19.5	pF
		g1 to g2 (cathode grounded)	---	---	Cout	6.1	6.7	pF
	Quality conformance inspection, part 3							
---	Life test	Group D F = 1050 ±50 Mhz; Ef = note 2; Eb = 5000 V dc; Ec2 = 1000 V dc; Ec1 = -200 V dc; pd = 1.5 kw; tp = 5-15 µs; Du = 0.01 (see note 4)	---	---	t	500	---	hours

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Method	Test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
	Quality conformance inspection, part 3 - continued							
---	Life test end points	Power output F = 405-450 MHz; Ef = 6.3; Eb = 5000 V dc; Ec2 = 1000 v; Ec1 = -50 V dc; pd = 2.2 kw; tp = 15-17 µs; Du = .005 (see note 4)	---	---	Po	12.0	---	kw

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17
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